

Product Requirements Document (PRD)

RankAutonomous — AI-Powered SEO & Link-Building SaaS

Product: RankAutonomous

Domain: rankautonomous.com

Product Type: SaaS Web Application

Primary Goal: Automate SEO research, content creation, website optimization, and backlink/mention-building workflows to help businesses increase organic search traffic.

1. Product Overview

RankAutonomous is an AI-powered SEO automation platform that allows users to connect their website and have the platform continuously analyze, research, create, optimize, and execute SEO activities.

The product should provide a simple experience: **Connect Website → AI Analyzes → AI Creates SEO Strategy → AI Produces Content → AI Assists With Link Building → Track Results**

The platform will operate on a subscription model: * **Monthly:** \$199/month * **Yearly:** \$149/month equivalent, billed annually * **Payment Gateway:** Stripe

The application should be designed as a scalable SaaS product rather than a simple marketing website.

2. Product Goals

Primary goals * Make SEO automation accessible to non-technical website owners. * Automatically analyze websites and identify SEO opportunities. * Generate SEO-focused content automatically. * Help users obtain backlinks and brand mentions. * Provide ongoing SEO recommendations and automation. * Track SEO performance and organic growth. * Support multiple website platforms/CMS types. * Provide a subscription-based SaaS business model.

Success criteria A user should be able to: * Create an account. * Select a subscription. * Connect a website. * Complete an initial SEO analysis. * See SEO issues and opportunities. * Configure their SEO strategy. * Generate articles. * Publish or export generated content. * Discover backlink opportunities. * Track SEO/traffic metrics. * Manage their subscription.

3. Target Users

Primary users * Small business owners * SaaS companies * E-commerce businesses * Bloggers * Marketing agencies * Website owners * Freelancers * SEO professionals

User characteristics The platform should assume that many users are not SEO experts. Therefore, complex SEO information should be translated into simple explanations such as: *"Your homepage is missing a meta description."* Instead of: *"Meta description optimization opportunity detected."*

4. User Roles

4.1 Customer Customers can: Register/login, Subscribe, Connect websites, Run SEO analysis, View SEO reports, Configure SEO settings, Generate articles, Manage articles, Manage backlink opportunities, View traffic/SEO performance, Manage integrations, Manage account, Manage billing.

4.2 Admin Admins can: View users, View websites, View subscriptions, Manage plans, Monitor AI usage, Monitor generated content, Monitor backlink activities, View system logs, Manage platform settings, Manage AI/API integrations, Suspend users, Manage support/contact requests.

5. Core Application Structure

Public Website Home, Features, How It Works, Pricing, FAQ, Contact, Login, Sign Up, Terms & Conditions, Privacy Policy.

Customer Application Dashboard, Website, SEO Analysis, SEO Strategy, Content, Backlinks, Keywords, Analytics, Recommendations, Integrations, Settings, Billing.

Admin Admin Dashboard, Users, Websites, Subscriptions, Content, SEO Activity, Backlinks, AI Usage, Integrations, System Logs, Settings.

6. Landing Page Requirements

The landing page should communicate the product immediately.

Hero section * **Headline:** "Put Your SEO on Autopilot." * **Supporting text:** RankAutonomous uses AI to analyze your website, create SEO content, discover link-building opportunities, and continuously improve your organic growth. * **CTA:** Start Growing → * **Secondary CTA:** See How It Works

Sections * **Hero:** Product headline, Description, CTA, Dashboard/product visual * **Problem:** Explain traditional SEO problems (Expensive SEO agencies, Manual keyword research, Time-consuming content creation, Difficult backlink acquisition, Constant technical SEO work) * **Solution:** Explain how RankAutonomous automates these processes. * **Features:** AI Website Analysis, AI Content Creation, AI Backlink & Mention Building, SEO Growth Automation. * **How it works:** Connect website → AI analyzes website → AI creates SEO strategy → AI executes SEO tasks → Monitor growth. * **Pricing:** Monthly (\$199/month) / Yearly (\$149/month equivalent). Show annual savings clearly. * **FAQ:** Website compatibility, AI content, integrations, subscriptions, cancellation, SEO results, publishing, backlinks.

7. Authentication

Users must be able to: Sign up, Login, Logout, Forgot password, Reset password, Email verification, Update profile, Change password. *Optional future support:* Google login, Microsoft login.

8. Onboarding Flow

After registration, users enter an onboarding wizard. * **Step 1 — Website:** "What website do you want to grow?" (e.g. <https://example.com>) * **Step 2 — Business information:** Business name, Industry, Country, Target audience, Business description. * **Step 3 — SEO goals:** Increase organic traffic, Rank for keywords, Generate leads, Increase sales, Improve authority, Increase brand visibility. * **Step 4 — Target locations:** Global, Country, State/region, City. * **Step 5 — Keywords:** Allow users to enter important keywords. AI should also suggest keywords automatically. * **Step 6 — Integration:** Ask how website is managed (WordPress, Shopify, Webflow, Custom, Other). * **Step 7 — Initial analysis:** Display progress (Crawling website, Analyzing technical SEO, Researching keywords, Analyzing competitors, Finding content opportunities, Preparing SEO strategy).

9. Dashboard

The dashboard should be the main control center. * **Dashboard cards:** Organic traffic, Organic keywords, Average ranking, Ranking changes, Backlinks, Referring domains, Published articles, SEO health score, Growth overview. * **Graphs:** Organic traffic, Keyword rankings, Backlinks, Domain authority. * **AI activity:** "AI published an article", "AI discovered 12 keyword opportunities", etc. * **Recommended actions:** e.g., High Priority: Your homepage has no optimized meta description. (CTA: Fix with AI).

10. Website Management

Users can add/manage websites. * **Website fields:** URL, name, Platform, Industry, Target country, Target audience, Primary keywords. * **Website statuses:** Connected, Analyzing, Active, Error, Disconnected. * **Multiple websites:** Architecture should support multiple websites per account in the future (limited by plan initially).

11. AI Website Analysis

- **Technical SEO analysis:** HTTPS, Page titles, Meta descriptions, Heading structure, Canonical URLs, Robots.txt, XML sitemap, Broken links, Redirects, Indexability, Duplicate content, Image alt text, Page speed indicators, Mobile friendliness indicators, Internal linking, URL structure.
- **On-page analysis:** Content quality, Keyword usage, Search intent, Heading optimization, Content depth, Internal links, Metadata, Image optimization.
- **AI analysis:** Identify content gaps, keyword opportunities, missing pages, optimization opportunities, topic opportunities, competitor opportunities.

12. SEO Health Score

Create an overall score (e.g., SEO Health: 78/100). * **Categories:** Technical SEO, On-page SEO, Content, Internal linking, Performance, Authority. * Each category receives: Score, Issues, Recommendations, Priority (Critical, High, Medium, Low).

13. AI SEO Strategy

Strategy should include: Target keywords, Topic clusters, Content plan, Internal linking strategy, Technical recommendations, Backlink strategy, Competitor opportunities. * **Example:** Primary Topic: Digital Marketing. Suggested content: What is Digital Marketing?, Digital Marketing Strategies, SEO vs PPC, etc.

14. Keyword Research

- **System provides:** Keyword, Search volume, Competition/difficulty, Search intent, Current ranking, Target URL, Opportunity score.
- **Categories:** Informational, Commercial, Transactional, Navigational.
- **AI discovery:** Related keywords, Long-tail keywords, Question keywords, Content opportunities, Keyword gaps.

15. Competitor Research

Users can enter competitors or allow AI to discover them. * **For each competitor:** Website, Ranking keywords, Content topics, Backlink opportunities, Content gaps, Strengths, Weaknesses. * AI should generate a competitor summary.

16. AI Content Generation

Automatically generate 1 SEO article per day. * **AI selects topic based on:** Target keywords, Search intent, Website niche, Existing content, Competitor research, Content gaps, SEO strategy. * **Article configuration:** Topic, Keyword, Word count, Tone, Language, Target audience, Location, CTA, Internal links, External references. * **Generated article includes:** SEO title, Meta description, URL slug, H1/H2/H3 headings, Article content, Suggested internal links, Suggested images, CTA.

17. Content Workflow

- **Statuses:** Idea → Draft → AI Review → User Review → Approved → Published
- **Actions:** Edit article, Regenerate section, Rewrite paragraph, Change tone, Improve SEO, Add keywords, Add links, Approve, Schedule, Publish, Export.

18. Automatic Publishing

Support publishing where the connected API allows it. * **WordPress:** Use REST API (Create/update post, media, categories, tags, featured image). * **Shopify & Webflow:** Where supported. * **Custom websites:** API integration, Webhook, CMS connector, Export, Manual publishing. * *Note:* Marketed as supporting multiple website platforms, rather than promising literal compatibility with every website.

19. Content Calendar

Calendar showing: Planned articles, Drafts, Approved articles, Published articles, Scheduled articles.

20. Backlink & Mention Building

Discover potential opportunities for: Guest posts, Relevant websites, Resource pages, Business directories, Industry publications, Brand mentions, Broken-link opportunities, Partnership opportunities. * **Data:** Website, URL, Domain info, Relevance, Opportunity type, Contact info (where publicly available), Status, Suggested action.

21. Backlink Campaign

Users can create campaigns (e.g., SaaS SEO Backlinks). Set targets (opportunities, niche, language, location).

* **Statuses:** Discovered, Qualified, Ready, Contacted, Replied, Accepted, Link acquired, Rejected.

22. AI Outreach Assistance

AI generates outreach messages personalized based on publicly available website information. * Users can edit, approve, copy, send through integrations, track status. Focus is on assisted and compliant outreach, not spam.

23. Brand Mention & Backlink Monitoring

- **Mentions:** New mentions, Brand references, New backlinks, Lost backlinks, Referring domains.
- **Backlinks:** Total backlinks, New/Lost backlinks, Referring domains, Anchor text, Target URL.

25. Internal Linking

AI analyzes existing pages and recommends internal links (Source page, Target page, Suggested anchor, Reason).

26. AI Recommendations

Centralized recommendations page. * **Data:** Title, Description, Impact, Difficulty, Priority, Suggested solution, AI action.

27. Analytics & SEO Reports

- **Metrics:** Organic traffic, Organic keywords, Ranking positions, Clicks, Impressions, CTR, Backlinks, Referring domains, Published content, SEO score. Date filters applied.
- **Reports:** Monthly automated reports (Health, traffic change, keywords, content published, backlinks, issues fixed, next month's strategy).

28. Integrations

- **Search Console:** Clicks, Impressions, CTR, Average position, Search queries.
- **Analytics:** Google Analytics (Sessions, Users, Landing pages, Conversions).
- **Websites:** WordPress, Shopify, Webflow, Custom/API.

31. Subscription & Billing

Use Stripe. Plans: Monthly (\$199), Annual (\$149/mo equivalent). * **Billing functionality:** Checkout, Subscription creation/renewal, Upgrade/Downgrade, Cancellation, Payment failure, Invoice history, Billing portal. Webhooks to sync statuses. Access restricted if inactive.

34. Settings

- **Account:** Name, Email, Profile.
- **Website:** URL, Business info, Target location, Industry, Keywords.
- **AI:** Tone, Language, Publishing preferences, Approval requirements.
- **Notifications:** Email notifications for content, backlinks, SEO issues, monthly reports, billing.

39. AI System & Automation Engine

- AI requests through backend API (Frontend → Backend → AI Provider). No keys exposed.
- **Background Jobs:** Daily (analyze changes, generate article, check backlinks), Weekly (audit, competitors, backlink monitoring), Monthly (SEO report).
- **Job Management:** Track status (Queued, Processing, Completed, Failed, Cancelled) and retries.

42. Database Structure

Core entities: users, websites, subscriptions, seo_audits, seo_issues, keywords, articles, backlink_opportunities, backlinks, integrations, analytics_snapshots, ai_jobs, notifications, audit_logs.

43. Security & Privacy

- HTTPS, Password hashing, API key protection, Input validation, Authorization checks, OAuth security, Stripe webhook validation.
- Privacy Policy, Terms & Conditions, clear data collection messaging.

45. SEO & UI/UX Requirements

- **Website:** Proper metadata, Open Graph, Schema, Sitemap, fast loading, responsive.

- **UI/UX:** Modern, AI, Premium SaaS feel. Clean dashboard, strong typography, data visualization, clear status indicators. Desktop-priority for data, mobile-usable for core actions.

48. Error Handling & Empty States

Clear errors for connection failures, AI failures, publishing, and payments. Empty states for missing data (e.g. "No website connected").

52. Recommended Tech Stack

- **Frontend:** React, TypeScript, Modern CSS
- **Backend:** Node.js, TypeScript, REST API
- **Database:** PostgreSQL
- **Auth/Payments:** Secure managed auth, Stripe
- **Hosting:** Client's Hostinger environment with external managed services as needed.

53. MVP Scope

- **Must Have:** Landing website, Auth, Onboarding, Website connection, Crawling, SEO audit & score, AI recommendations, Keyword research, AI content generation (Daily article), CMS integration (1+), Backlink discovery/tracking, Dashboard, Analytics integration, Stripe subscriptions, Admin panel.
- **Phase 2:** Advanced integrations, automated outreach, agency accounts, white-label reports.

55. SEO/Backlink Automation Safety

Avoid automated spam or deceptive link schemes. Prioritize relevant websites, legitimate outreach, and useful content.

56. Core User Journey

LANDING PAGE → SIGN UP → CHOOSE PLAN → STRIPE CHECKOUT → ONBOARDING → CONNECT WEBSITE → WEBSITE ANALYSIS → SEO SCORE → AI SEO STRATEGY → KEYWORD RESEARCH → CONTENT PLAN → DAILY AI ARTICLE → USER APPROVAL → PUBLISH → BACKLINK OPPORTUNITIES → SEO/TRAFFIC MONITORING → MONTHLY REPORT → CONTINUOUS AUTOMATION

57. Definition of Done

Platform is fully functional end-to-end, users can subscribe securely, connect sites, generate content, publish to integrations, discover backlinks, view analytics, and admin can manage the ecosystem.

58. Final Product Vision

RankAutonomous feels like having an AI SEO team working continuously in the background across four pillars: **ANALYZE, CREATE, BUILD, GROW.**